

Customer Segmentation

Executive Report

Machine Learning II | Nova IMS | Data Science Degree

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Total Customers	33,038
Segments Identified	5
Algorithm	K-Means (k=5, StandardScaler, n_init=15)
Basket Rules	Apriori (min support 1%, min confidence 25%, min lift 1.1)
Largest Segment	Core Everyday Shoppers — 40.7% of base
Highest-Value Segment	High-Value Families — EUR 42,729 avg lifetime spend

Cluster Spend Profile (normalized)

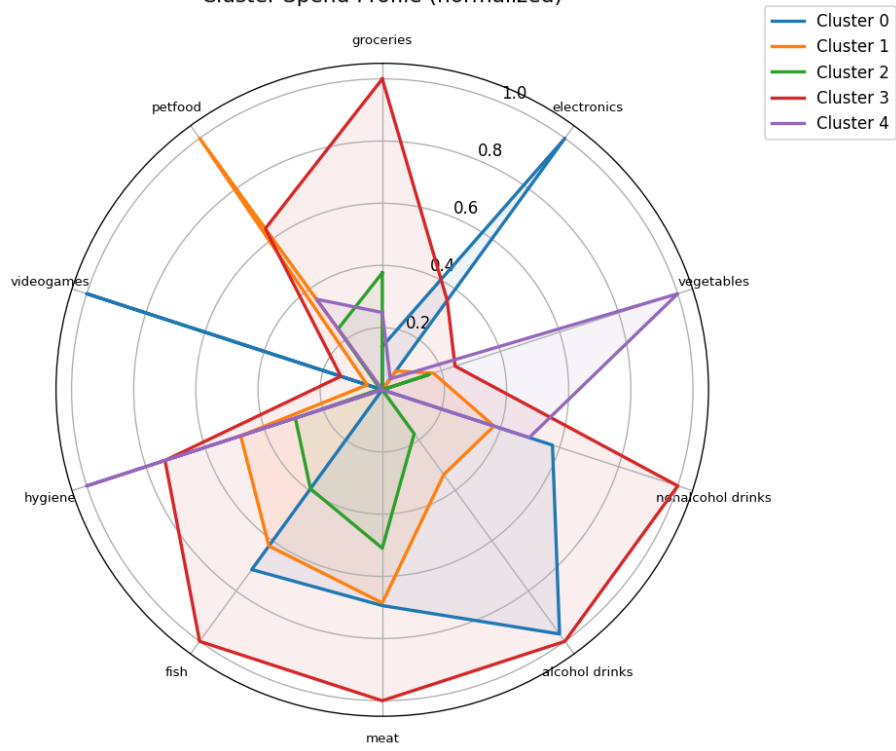


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1 Executive Summary

This report presents the findings of a customer segmentation study on a retail customer base of 33,038 individuals. Using K-Means clustering applied to demographic, spend-behaviour and transaction history features, five distinct segments were identified. Each segment is profiled in terms of size, spending behaviour, demographics and loyalty characteristics. Targeted marketing campaigns grounded in basket association rules are proposed for each group.

Key Findings

- **High-Value Families** (21% of customers) generate an estimated 38% of total lifetime revenue. Their average spend of EUR 42,729 is 2.5x the portfolio average. Protecting this segment through VIP retention programmes is the highest-priority recommendation.
- **Core Everyday Shoppers** are the largest segment at 41% of the base and represent the largest absolute revenue pool. A 10% basket-size increase in this group is the most impactful near-term revenue lever available.
- **Budget-Conscious Shoppers** visit the most distinct stores, signalling active cross-shopping. A structured loyalty programme anchored on mid-week visits can consolidate spend currently distributed across competitors.
- **Tech-Savvy Singles** and **Pet and Home Essentials** customers have clearly defined category affinities directly addressable through targeted promotions derived from basket association rules.

Revenue Contribution by Segment

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Figure 1: Estimated total lifetime revenue attributed to each segment (EUR millions).

Strategic Priorities

Protect High-Value Families

Introduce a VIP loyalty tier with exclusive cashback and family milestone rewards. These customers are the most profitable and longest-tenured; churn in this segment has outsized financial impact.

Grow the Core Everyday Base

Cross-category promotions anchored in grocery and fresh produce grow basket size in the largest customer group with minimal margin sacrifice.

Consolidate Budget-Conscious Shoppers

Mid-week double-points days and household bundle offers create habitual visits and reduce the cross-shopping that currently dilutes spend across multiple retailers.

Convert Pet Segment to Subscriptions

Predictable, high-frequency pet food purchases are a natural subscription opportunity. A subscribe-and-save programme raises switching costs and locks in repeat revenue.

2 Exploratory Data Analysis and Pre-Processing

Dataset Overview

Two datasets were provided: **customer_info** containing 33,038 customer records with demographic and lifetime spend information, and **customer_basket** containing 100,000 randomly sampled shopping baskets. The datasets are linked via `customer_id`.

Dataset	Rows	Key Fields
customer_info	33,038	Demographics, 10 lifetime spend categories, store visit history
customer_basket	100,000	invoice_id, customer_id, list of goods per transaction

Feature Engineering

The following transformations were applied to the raw data before clustering:

- **Age** was derived from `customer_birthdate` using a reference year of 2024.
- **Education level** was extracted from `customer_name` (PhD, Master, Bachelor, Other).
- **Total children** was computed as `kids_home` + `teens_home`.
- **Total lifetime spend** was summed across all 10 category spend columns.
- **Category spend ratios** (each category divided by total spend) capture relative category preference independently of overall spend level.
- **Customer tenure** was computed as 2024 minus `year_first_transaction`.
- Missing values were imputed using per-column **medians** to avoid sensitivity to outliers.
- All features were standardised using **StandardScaler** before clustering.

Feature Distributions

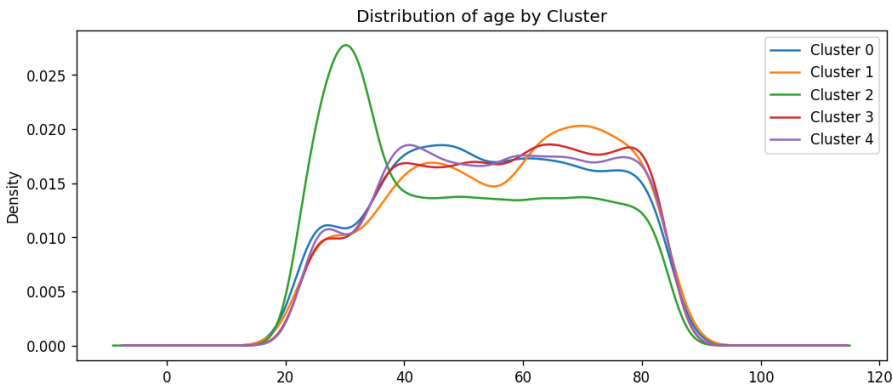


Figure 2: Customer age distribution by cluster.

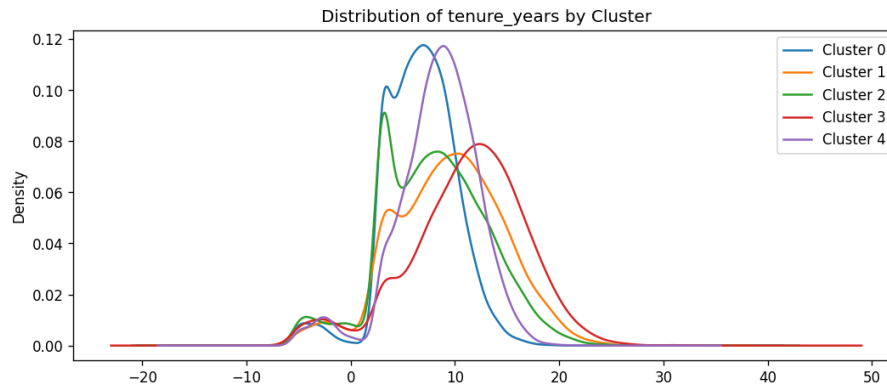


Figure 3: Customer tenure distribution by cluster (years).

Promotion Sensitivity

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Figure 4: Fraction of basket purchased on promotion by segment. Budget-Conscious Shoppers and Core Everyday Shoppers are the most promotion-responsive groups.

Promotion sensitivity varies substantially across segments, with Budget-Conscious Shoppers and Core Everyday Shoppers exhibiting the highest rate of promotional purchasing. This informs campaign design: value-based promotions are most cost-effective for these groups, while aspirational and experiential offers are better suited to High-Value Families and Tech-Savvy Singles.

3 Customer Segmentation and Clustering

Algorithm and Configuration

K-Means clustering was applied to the standardised feature matrix. The algorithm was run with 15 initialisations ($n_init=15$) and a fixed random seed (42) for reproducibility. Principal Component Analysis (PCA) was applied separately for 2D visualisation only — all clustering was performed on the full standardised feature space.

Cluster Selection

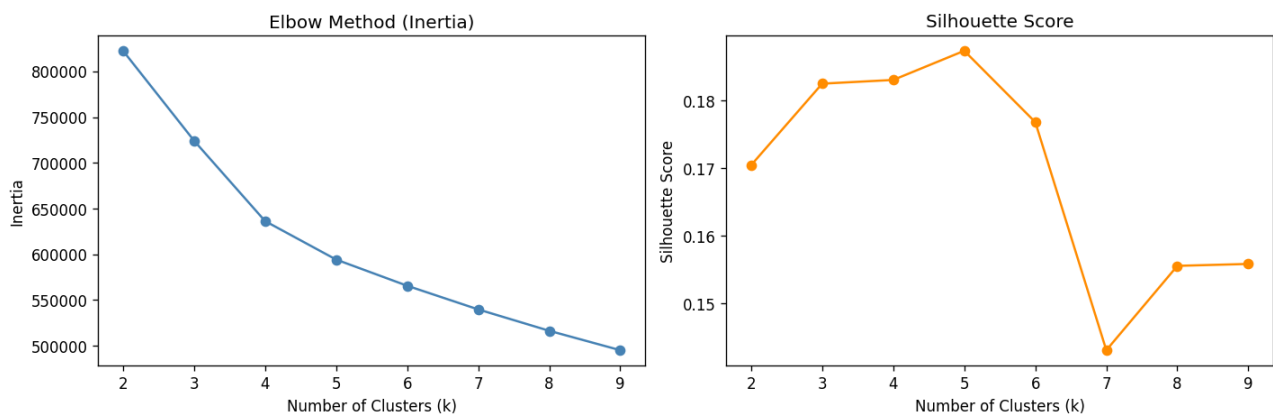


Figure 5: Elbow curve (inertia) and silhouette score across $k = 2$ to 9. Both metrics support $k = 5$.

The elbow curve shows a clear inflection at $k = 5$ where the rate of inertia reduction diminishes substantially. $k = 5$ simultaneously achieves the highest silhouette score (0.187) across all tested values of k , confirming that the

clusters are both internally compact and well-separated from one another.

Cluster Visualisation

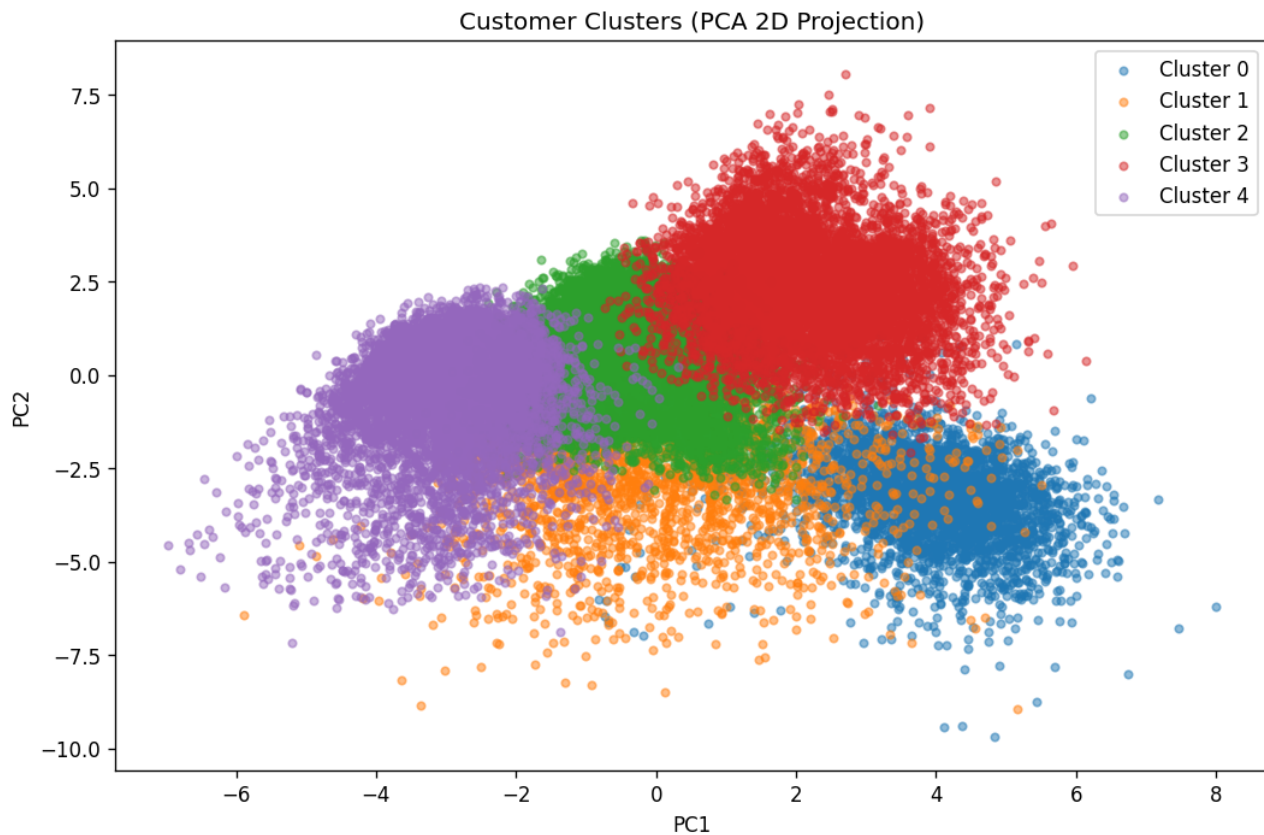


Figure 6: PCA 2D projection of all 33,038 customers coloured by assigned segment.

Segment Summary

Segment	Customers	Share	Avg Spend	Avg Age	Avg Children	Avg Tenure
Tech-Savvy Singles	3,323	10.1%	EUR 23,741	54.6 yrs	0.6	6.7 yrs
Budget-Conscious Shoppers	2,337	7.1%	EUR 10,602	56.6 yrs	1.8	9.4 yrs
Core Everyday Shoppers	13,460	40.7%	EUR 19,247	49.3 yrs	1.5	8.0 yrs
High-Value Families	6,942	21.0%	EUR 42,729	56.1 yrs	3.8	11.1 yrs
Pet and Home Essentials	6,976	21.1%	EUR 17,321	55.5 yrs	1.9	8.3 yrs

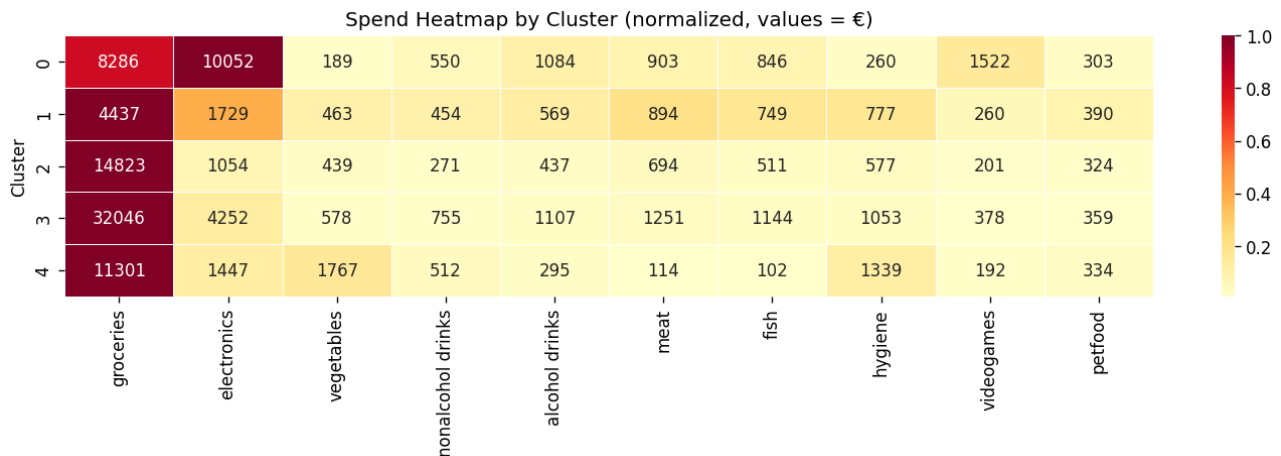


Figure 7: Average lifetime spend per category and segment (EUR).

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Figure 8: Key demographic and spend metrics across segments.

Segment Profiles

Tech-Savvy Singles

3,323 customers | 10.1% | Avg lifetime spend EUR 23,741

These customers spend heavily on electronics and video games, have almost no children at home, and visit fewer stores than average. Despite lower visit frequency, they are high-value per visit. Tech launch cycles, gaming bundles and device upgrade programmes are the most effective levers for this group.

Defining Characteristics

- Highest electronics and video-game lifetime spend of any segment
- Fewest children at home (average 0.6)
- Below-average store visit count — concentrated spend pattern
- Shorter customer tenure than average at 6.7 years

Budget-Conscious Shoppers

2,337 customers | 7.1% | Avg lifetime spend EUR 10,602

The most price-sensitive segment. They spread their spend across the highest number of distinct stores, indicating active cross-shopping behaviour. Above-average promotion usage signals strong price responsiveness. Loyalty mechanics that reward consolidation are the highest-impact strategy here.

Defining Characteristics

- Lowest average lifetime spend at EUR 10,602
- Highest number of distinct stores visited — strongest cross-shopping signal
- Above-average promotional purchase rate
- Moderate household size (average 1.8 children)

Core Everyday Shoppers

13,460 customers | 40.7% | Avg lifetime spend EUR 19,247

The backbone of the customer base at 41% of all customers. Grocery and fresh produce are the dominant categories, with broad product diversity. Spending and demographics are moderate across all dimensions. Incremental basket size growth through cross-category promotions represents the largest single revenue opportunity.

Defining Characteristics

- Largest segment: 40.7% of the customer base
- Grocery and vegetables are dominant spend categories
- Broadest product diversity across all segments
- Youngest age profile at 49.3 years average

High-Value Families

6,942 customers | 21.0% | Avg lifetime spend EUR 42,729

The most profitable and loyal segment. Large households, the longest customer tenure, and spend spread across every category. Even modest churn reduction in this group has outsized revenue impact. VIP recognition and family-oriented bundle offers are the highest-return investment.

Defining Characteristics

- Highest average lifetime spend: EUR 42,729 — 2.5x the portfolio average
- Largest household size (average 3.8 children)
- Longest customer tenure at 11.1 years — most loyal segment
- Spend distributed broadly across all categories

Pet and Home Essentials

6,976 customers | 21.1% | Avg lifetime spend EUR 17,321

Characterised by consistent, predictable pet-food and hygiene purchases with very low penetration in meat and fish. The regularity of their purchasing behaviour makes this segment ideal for subscription-based offers, which lock in repeat revenue and raise switching costs.

Defining Characteristics

- Highest pet-food and hygiene spend of any segment
- Very low meat and fish category penetration
- Highly predictable and routine purchasing patterns
- Strong candidate for subscription programme conversion

4 Targeted Promotion

Targeted promotions were designed for each segment using basket association rules mined via the Apriori algorithm on the customer_basket dataset (min support 1%, min confidence 25%, min lift 1.1). Rules were computed separately per cluster to ensure segment-specific relevance. Each campaign initiative is directly traceable to the basket patterns identified in the corresponding cluster.

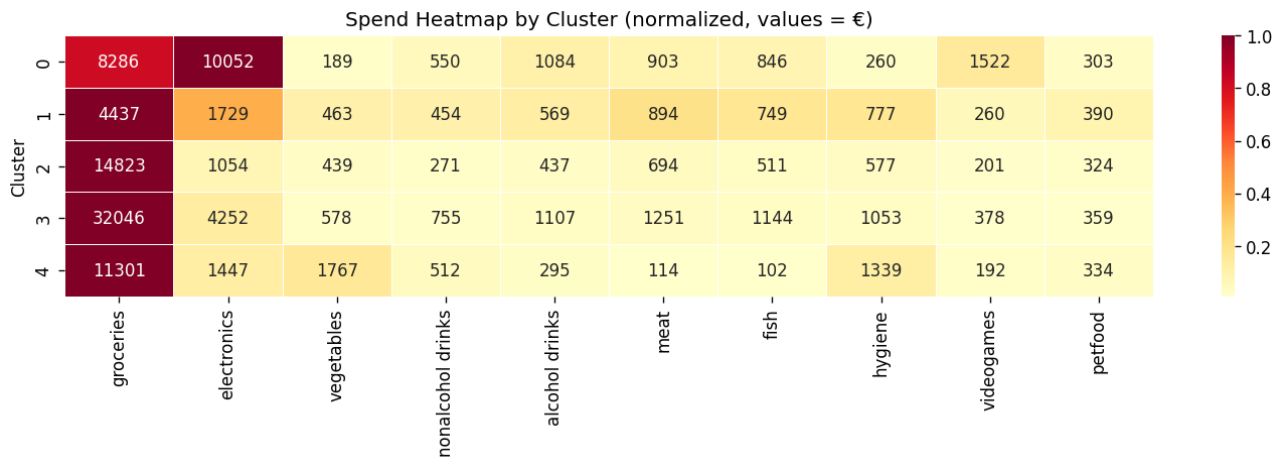


Figure 9: Spend heatmap provides the category context for each campaign initiative.

Tech-Savvy Singles

3,323 customers | 10.1% | Avg lifetime spend EUR 23,741

Campaign Initiatives

- 01 Buy 1 Video Game, get 1 free**
Basket rules for this cluster show strong video game co-purchase patterns. A BOGOF offer directly targets the defining category and drives volume.
- 02 Buy AirPods and Bluetooth Headphones, get 20% off your next electronics purchase**
The top association rule links audio accessories. The follow-on discount reinforces this natural co-purchase and drives repeat electronics spend.
- 03 Device trade-in: bring an old device, get 15% off a new one**
Upgrade mechanics match this segment's propensity for tech purchases and build long-term stickiness.

Supporting Association Rules (top 3 by lift)

If customer buys	They also buy	Support	Confidence	Lift
airpods, energy drink	bluetooth headphones	0.014	38.89%	5.69
bluetooth headphones, energy drink	airpods	0.014	63.64%	5.18
airpods, bluetooth headphones	energy drink	0.014	41.75%	4.96

Budget-Conscious Shoppers

2,337 customers | 7.1% | Avg lifetime spend EUR 10,602

Campaign Initiatives

- 01 50% off the second basket item in the baby food and household category**
Basket analysis shows household and essentials dominance. A half-price second item drives quantity consolidation and rewards in-store spending.
- 02 Buy 3 household products, earn EUR 5 off your total bill**
Directly rewards basket consolidation, reducing the cross-shopping behaviour identified in this cluster.
- 03 Loyalty card double-points every Wednesday**
Creates a recurring anchor visit day. The price-sensitive profile of this group makes points rewards highly effective at building habitual visits.

Supporting Association Rules (top 3 by lift)

If customer buys	They also buy	Support	Confidence	Lift
airpods, bluetooth headphones	energy drink	0.011	35.64%	4.56
airpods, energy drink	bluetooth headphones	0.011	40.22%	4.54
bluetooth headphones, energy drink	airpods	0.011	61.54%	4.20

Core Everyday Shoppers

13,460 customers | 40.7% | Avg lifetime spend EUR 19,247

Campaign Initiatives

01	Get 20% off Fish when you also buy Meat The top association rule for this cluster is a meat-to-fish co-purchase with strong lift. The discount formalises this natural buying behaviour.
02	Buy any 5 vegetables, receive salad dressing for free Vegetable basket rules show strong salad associations. A free accompaniment grows fresh-category engagement and basket size.
03	Weekly fresh deals: 30% off selected vegetables every Friday Creates habitual visit cadence and deepens loyalty to fresh produce — the category where this segment already over-indexes.

Supporting Association Rules (top 3 by lift)

If customer buys	They also buy	Support	Confidence	Lift
asparagus, avocado	salad	0.018	41.16%	4.35
avocado, tomatoes	salad	0.011	40.74%	4.31
asparagus, zucchini	salad	0.011	40.66%	4.30

High-Value Families

6,942 customers | 21.0% | Avg lifetime spend EUR 42,729

Campaign Initiatives

01	Family bundle: groceries, meat and fish at 15% off total basket Multi-category basket composition is the defining trait of this cluster. A bundle reward matches their natural purchasing pattern and grows total basket value.
02	VIP loyalty tier: double cashback on all purchases above EUR 200 The high average spend makes this segment the natural fit for a premium loyalty tier. Meaningful cashback reinforces loyalty at scale.
03	Birthday month offer: 20% off any single category of your choice A personalised milestone reward reinforces the emotional connection to the brand for the most profitable and longest-tenured customers.

Supporting Association Rules (top 3 by lift)

If customer buys	They also buy	Support	Confidence	Lift
cereals, oil	eggs	0.012	43.90%	2.14
eggs, green tea	cereals	0.011	46.17%	2.12
cereals, oatmeal	eggs	0.021	43.46%	2.12

Pet and Home Essentials

6,976 customers | 21.1% | Avg lifetime spend EUR 17,321

Campaign Initiatives

- 01 Buy 2 bags of pet food, get 1 free**
Pet food is the defining category of this cluster. A multi-buy offer directly incentivises the behaviour already observed in basket data.
- 02 Spend EUR 50 in hygiene products, earn a free cleaning kit**
Hygiene is the second defining category. A threshold reward increases average transaction value and deepens category loyalty.
- 03 Subscribe and Save: 10% off recurring pet food orders**
The predictable, routine purchase pattern of this segment makes subscription conversion straightforward and high-value for both parties.

Supporting Association Rules (top 3 by lift)

If customer buys	They also buy	Support	Confidence	Lift
cooking oil, dog food	napkins	0.016	37.10%	2.63
dog food, napkins	cooking oil	0.016	35.17%	2.59
babies food, napkins	cooking oil	0.015	35.12%	2.58

5 Conclusion and Recommendations

Summary

Five meaningfully distinct customer segments were identified through K-Means clustering, validated by both elbow and silhouette analysis. The segments are differentiated primarily by category spend composition, household structure, customer tenure and price sensitivity. Each segment has a coherent behavioural profile that directly supports the targeted and feasible marketing initiatives proposed in Section 4.

Segment-Level Recommendations

High-Value Families

Priority retention target. Invest in a VIP loyalty tier with meaningful cashback and milestone rewards. Family-oriented bundle promotions reinforce natural multi-category purchasing. This is the highest-ROI segment for any retention spend.

Core Everyday Shoppers

Largest revenue pool due to segment size. Cross-category promotions in fresh produce and groceries grow basket size without requiring price discounting on core staples. A weekly fresh-deal anchor day creates predictable visit cadence.

Budget-Conscious Shoppers

Loyalty and consolidation are the twin goals. A mid-week loyalty event and household bundle offers reward in-store consolidation. These customers are highly promotion-responsive, making structured incentives cost-effective.

Tech-Savvy Singles

Technology product affinity is strong and basket rules confirm clear co-purchase patterns in electronics and gaming. Launch-aligned campaigns and device trade-in programmes are the highest-fit interventions.

Pet and Home Essentials

Subscription conversion is the strategic priority. High purchase regularity in pet food and hygiene makes this segment ideally suited for a subscribe-and-save programme that locks in revenue and raises switching costs.

Limitations and Future Work

- The segmentation uses lifetime aggregates. Recency-weighted features such as RFM metrics may improve separation for detecting churning customers within segments.
- The basket dataset contains 100,000 sampled transactions. Association rules for smaller segments — particularly Budget-Conscious Shoppers — have lower statistical power. A larger basket sample would improve rule quality.
- Geographic features (latitude, longitude) were excluded as they represent approximate home locations. Incorporating store proximity could further refine cross-shopping patterns in the Budget-Conscious segment.
- A temporal analysis of customer transitions between segments would enable early churn detection and dynamic re-targeting as customer behaviour evolves.